Math 242, S09 Name: Answer

Quiz 2 SSN: xxx-xx-

Instructions: There are totally 1 problem. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (10 points) Find a general solution for each of the following first-order differential equations.

(a) $\frac{dy}{dx} = \frac{1}{\sqrt{2x+1}}$.

$$y = \int \frac{1}{\sqrt{2x+1}}\, dx + C \overset{u=2x+1}{\underset{du=2dx}{=}} \int \frac{1}{\sqrt{u}} \frac{1}{2} du + C$$

$$= \sqrt{u} + C$$

$$= \sqrt{2x+1} + C$$

(b) $\frac{dy}{dx} = x \sin x$.

$$y = \int x \sin x\, dx + C$$

$$= \int x (-\cos x)'\, dx + C$$

$$= -x\cos x - \int (-\cos x)\, dx + C$$

$$= -x\cos x + \sin x$$